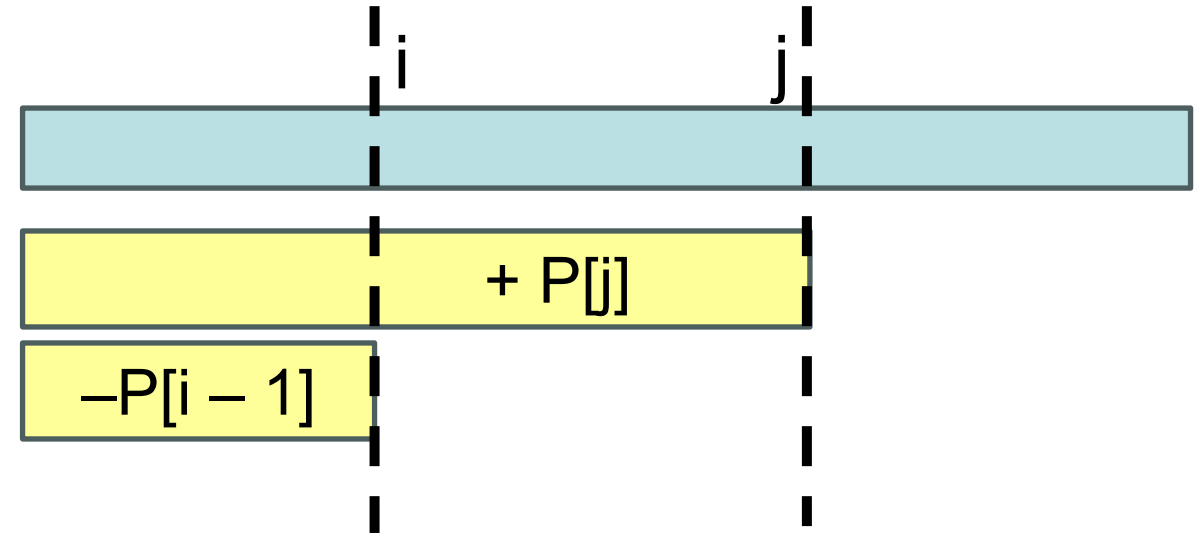
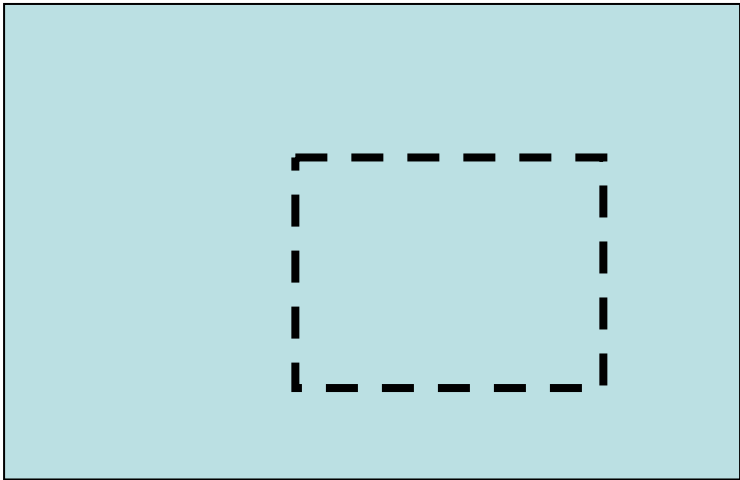


Five Nice Ideas...

4. Range = Prefix – Prefix

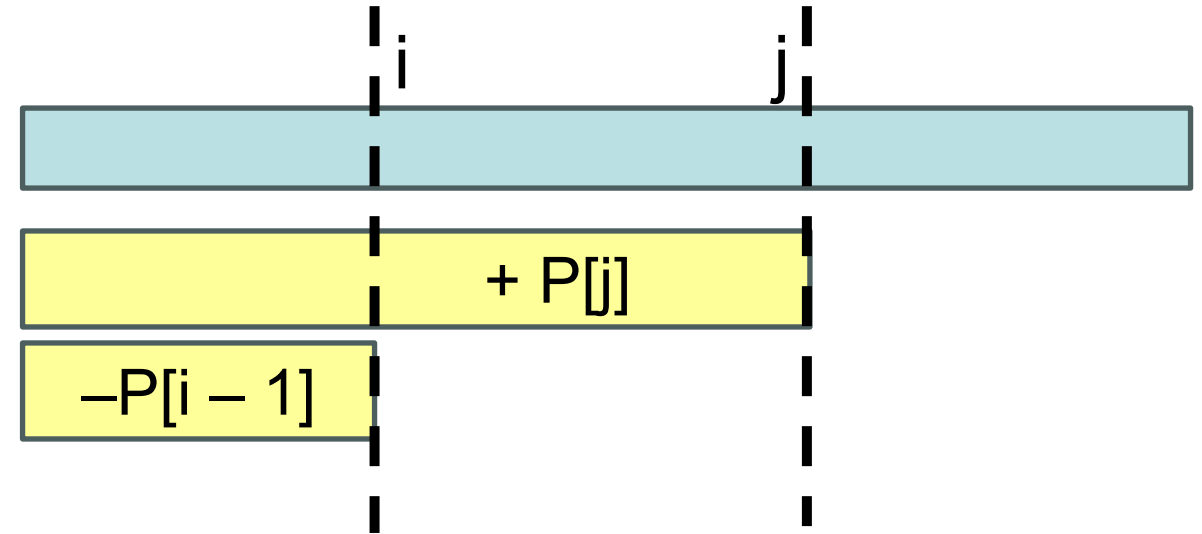
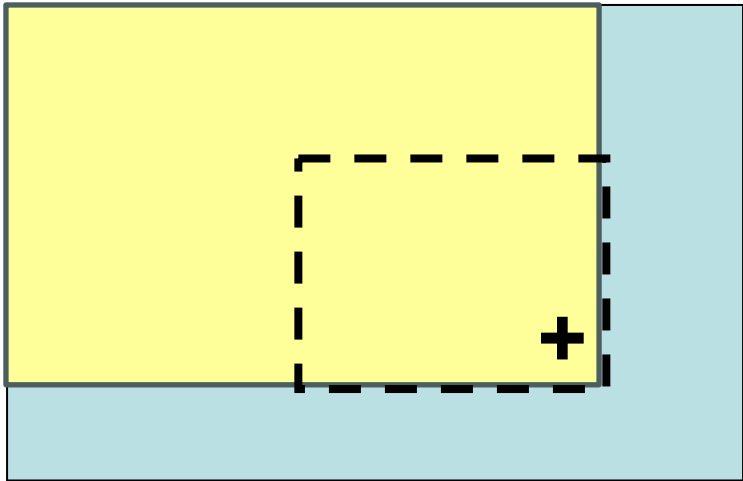
- Calculate in $O(1)$ time the sum of any range in a static array.
- ... and also any rectangle sum in a static 2D array.



Five Nice Ideas...

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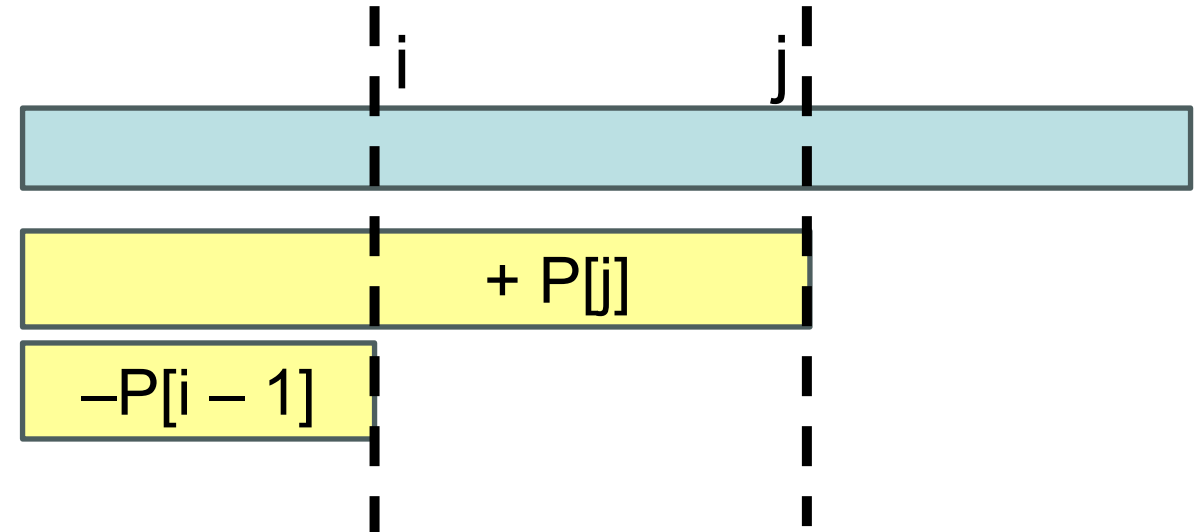
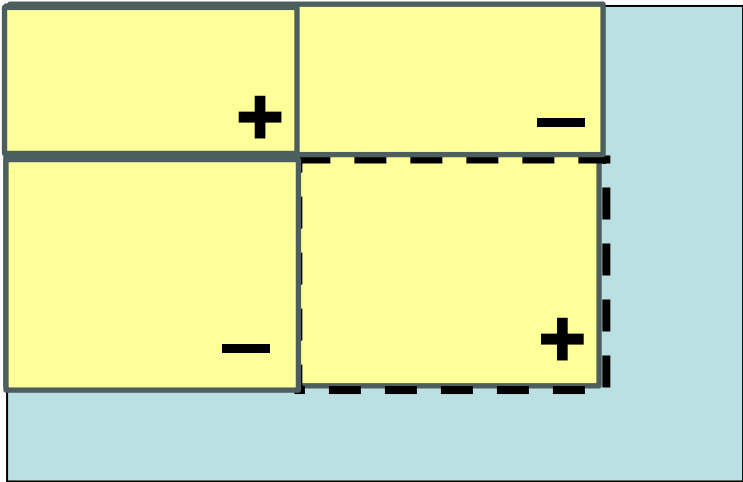


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Five Nice Ideas...

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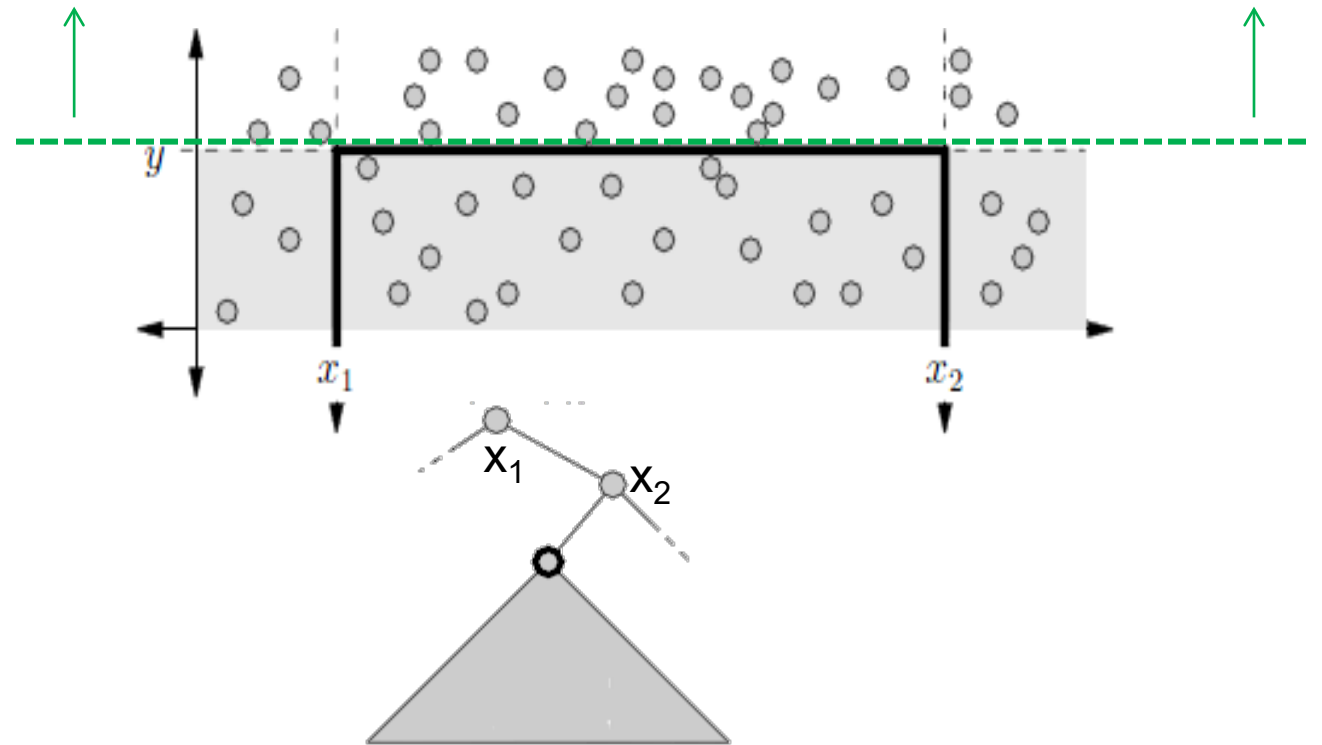


- “Prefix sums” here correspond to “upper left quadrant” sums.

Five Nice Ideas...

4. Range = Prefix – Prefix

- Answer “3-sided” counting queries in 2D.
- Offline setting = queries known in advance.
- A sweep line approach will do the trick here, running in $O((n + q) \log n)$ total time if n points and q queries.
- Persistence (our enrichment lecture) yields $O(\log n)$ query time also in an online setting!



Points with x coordinates in range (x_1, x_2) that have been swept so far

Five Nice Ideas...

4. Range = Prefix – Prefix

- What about 4-sided counting queries?
- Just a difference of two “prefix sum” 3-sided queries!

